B

Rosenblatt and Other Transformations

B.1 Rosenblatt Transformation

A dependent random vector $\mathbf{X} = \{X_1, X_2, \ldots, X_n\}$ may be transformed to the independent Uniform distributed random vector $\mathbf{R} = \{R_1, \ldots, R_n\}$ through the Rosenblatt (1952) transformation $\mathbf{R} = T\,\mathbf{X}$ given by

$$\begin{aligned}
r_1 &= P(X_1 \leq x_1) = F_1(x_1) \\
r_2 &= P(X_2 \leq x_2 | X_1 = x_1) = F_2(x_2 | x_1) \\
&\cdot \\
&\cdot \\
&\cdot \\
r_n &= P(X_n \leq x_n | X_1 = x_1, \ldots, X_{n-1} = x_{n-1}) = F_n(x_n | x_1, \ldots, x_{n-1})
\end{aligned} \tag{B.1}$$

where $F_i()$ is shorthand for the conditional cumulative distribution function $F_{X_i|X_{i-1},\ldots,X_1}(\;)$.

If the joint probability density function $f_{\mathbf{X}}(\;)$ is known, then $F_i(\;)$ can be determined as follows. From Section A.6.2, the conditional probability density function $f_i(\;)$ is given by

$$f_i(x_i | x_1, \ldots, x_{n-1}) = \frac{f_{\mathbf{X}_i}(x_1, \ldots, x_i)}{f_{\mathbf{X}_{i-1}}(x_1, \ldots, x_{i-1})} \tag{B.2}$$

where $f_{\mathbf{X}_j}(x_1, \ldots, x_j)$ is a marginal probability density function, obtained from

$$f_{\mathbf{X}_j}(x_1, \ldots, x_j) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f_{\mathbf{X}}(x_1, \ldots, x_n) dx_{j+1}, \ldots, dx_n \tag{B.3}$$

$F_i(\;)$ is then obtained by integrating $f_i(\;)$ given by (B.2) over x_i:

$$F_i(x_i | x_1, \ldots, x_{n-1}) = \frac{\int_{-\infty}^{\infty} f_{\mathbf{X}_i}(x_1, \ldots, x_{i-1}, t)\, dt}{f_{\mathbf{X}_{i-1}}(x_1, \ldots, x_{i-1})} \tag{B.4}$$

Structural Reliability Analysis and Prediction, Third Edition. Robert E. Melchers and André T. Beck.

With all the conditional cumulative distribution functions $F_i(\ )$ determined in this way, (B.1) may be inverted successively to obtain

$$\begin{aligned} x_1 &= F_1^{-1}(r_1) \\ x_2 &= F_2^{-1}(r_2|x_1) \\ &\cdot \\ x_n &= F_n^{-1}(r_n|x_1, \dots, x_{n-1}) \end{aligned} \tag{B.5}$$

It follows immediately that (B.5) can be used to generate the random vector **X** with probability density function $f_{\mathbf{X}}(\ )$ from **R**. A practical difficulty is that, unless $F_i(\ )$ is simple in form, the inversion will need to be done numerically.

As noted by Rosenblatt (1952), there are $n!$ possible ways in which expressions (B.1) can be written, depending on the numbering adopted for the variables in **X**. Equally, of course, there are $n!$ possible ways of conditioning the X_i in expression (B.1), as seen for the simple case $n = 2$ [Rubinstein, 1981]:

$$F_{X_1X_2}(x_1, x_2) = F_{X_1}(x_1) f_{X_2|X_1}(x_2|x_1) = F_{X_2}(x_2) f_{X_1|X_2}(x_1|x_2)$$

It is probably obvious that this freedom can lead to considerable differences in the difficulty of solving for **X**, i.e. in solving (B.5).

Unfortunately, the above method is not always useful for practical problems since $F_{\mathbf{X}}(\ )$, or the conditional probability density functions $f_i(\ )$ which characterize the dependence structure of the problem, are not always known. More commonly only some estimate of correlation may be available from the data. This case is discussed in Section B.2 below. A special case arises when the X_i are independent. All the conditional requirements in (B.5) then disappear and each transformation takes the form $x_i = F_i^{-1}(r_i)$ independent of all other $x_j (j \neq i)$.

The Rosenblatt transformation may be used to transform from one distribution to another by applying (B.1) twice, using **R** as a transmitter, e.g.

$$\begin{aligned} F_1(u_1) &= r_1 = F_1(x_1) \\ F_2(u_2|u_1) &= r_2 = F_2(x_2|x_1) \\ &\text{etc.} \end{aligned} \tag{B.6}$$

A particular case of interest is where **U** in (B.6) is standard Normal distributed, with **X**, say, a vector of correlated random variables and **U** uncorrelated (independent). Then (B.6) may be written as

$$\begin{aligned} x_1 &= F_1^{-1}[\Phi(u_1)] \\ x_2 &= F_2^{-1}[\Phi(u_2)|x_1] \\ &\text{etc.} \end{aligned} \tag{B.7}$$

In practice, solution of (B.7) requires multiple integration [*cf*. (B.4)]. This technique is used in Section 4.4.3.1 to convert non-Normal distributed random variables to equivalent Normal random variables. Where both **U** and **X** are Normal vectors, an easier approach to finding the transformation implied by (B.7) is to make direct use of the special properties of the Normal distribution. This is considered in Section B.3.

This transformation usually is attributed to Rosenblatt (1952) but appears to have been given earlier by Segal (1938). It was first suggested for use in structural reliability by Hohenbichler and Rackwitz (1981).

B.2 Nataf Transformation

A dependent random vector $\mathbf{X} = (X_1, \dots, X_n)$, for which the marginal cumulative distribution functions $F_{X_i}(\ ), i = 1, \dots, n$ and the correlation matrix $\mathbf{P} = \{\rho_{ij}\}$ are known, may have assigned to it an approximate but completely specified joint probability distribution function $F_{\mathbf{X}}(\mathbf{x})$. It also may be transformed to the standardized Normal random variables $\mathbf{Y} = (Y_1, \dots, Y_n)$ in $\mathbf{y}$ space, given by

$$Y_i = \Phi^{-1}[F_{X_i}(X_i)], \qquad i = 1, \dots, n \tag{B.8}$$

where $\Phi(\ )$ is the standard Normal cumulative distribution function. Let $\mathbf{Y} = (Y_1, \dots, Y_n)$ be n-dimensional standard Normal with joint probability density function $\phi_n(\mathbf{y}, \mathbf{P}')$ having zero means, unit standard deviations and correlation matrix $\mathbf{P}' = \{\rho'_{ij}\}$. Then, with the usual rules for transformation of random variables, the approximate joint density function $f_{\mathbf{X}}(\ )$ in $\mathbf{x}$ space is (Nataf, 1962):

$$f_{\mathbf{X}}(\mathbf{x}) = \phi_n(\mathbf{y}, \mathbf{P}') \cdot |\mathbf{J}| \tag{B.9a}$$

with

$$|\mathbf{J}| = \frac{\partial(y_1, \dots, y_n)}{\partial(x_1, \dots, x_n)} = \frac{f_{X_1}(x_1).f_{X_2}(x_2) \dots f_{X_n}(x_n)}{\phi(y_1)\phi(y_2) \dots \phi(y_n)} \tag{B.9b}$$

To solve for $\mathbf{P}' = \{\rho'_{ij}\}$ in (B.9) consider any two random variables (X_i, X_j) and the correlation between them as:

$$\rho_{ij} = \frac{\mathrm{cov}[X_i X_j]}{\sigma_{X_i}\sigma_{X_i}} = E[Z_i Z_j] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} z_i z_j \phi_2(y_i, y_j;\ \rho'_{ij})\, dy_i dy_j \tag{B.10}$$

with $Z_i = (X_i - \mu_{X_i})/\sigma_{X_i}$ and with y_i and y_j dummy variables. Here the correlation matrix $\mathbf{P}' = \{\rho'_{ij}\}$ can be obtained from the known $\mathbf{P} = \{\rho_{ij}\}$ iteratively from (B.10). This is readily programmed but is tedious to solve since the unknown is contained within the double integral. Empirical, approximating expressions for the ratio $R = \rho'_{ij}/\rho_{ij}$ are given in Tables B.1 to B.3 below for a selected set of distributions for random variables. A more complete set of approximating expressions is available in the literature (Liu and Der Kiureghian, 1986; Der Kiureghian and Liu, 1986).

Once the $\mathbf{P}' = \{\rho'_{ij}\}$ are obtained for any pair (X_i, X_j) expression (B.8) may be used to obtain the correlated standard Normal distribution in $\mathbf{y}$ space. For the two variables involved, an orthogonal transformation (see below) may be used to obtain independent standard Normal random variables for use in FOSM theory.

The Nataf transformation (B.9) relies on the following fundamental properties being valid for expression (B.10) [Liu and Der Kiureghian, 1986]:

(1) ρ_{ij} is an increasing function of ρ'_{ij};

(2) $\rho'_{ij} = 0$ for $\rho_{ij} = 0$ and vice versa;

(3) $R \geq 1$;
(4) If both marginal distributions are Normal, $R = 1$;
(5) If one of the marginal distributions is Normal, R is a constant;
(6) R is invariant to increasing linear transformations of X_i and X_j;
(7) R is independent of the parameters of the marginal distributions of Type 1 (those whose two-parameter distributions can be reduced to a parameter-free form by a linear transformation);
(8) R is a function of the coefficient of variation $V = \sigma / \mu$ for marginal distributions of Type 2 (those that cannot be reduced to a parameter-free form by a linear transformation).

Typical examples of the Type 1 and Type 2 distributions with cross-reference to their descriptions in Appendix A are given in Table B.1.

The approximate expressions for $R = \rho'_{ij} / \rho_{ij}$ are based on a polynomial of the form

$$R = a + bV_i + cV_i^2 + d\rho + e\rho^2 + f\rho V_i + gV_j + hV_j^2 + k\rho V_j + lV_iV_j$$

with the coefficients given in Tables B.2 and B.3. The maximum error in the approximations, given in Table B.2, generally is much below 1%. The exception is when the Exponential distribution is involved, in which case the maximum error can be up to about 2% or more when there is high negative correlation. This tends to be due to the shape of the Exponential distribution being considerably different from the Normal distribution used as the basis for the method. The maximum errors for the approximations in the formulae given in Table B.3 are shown in the last column of the table.

Table B.1 Selected two-parameters distributions.

Type & name	Reference (Appendix A)	Symbol	Standardized form
Type 1			
Normal	A 5.7	N	$\Phi(y)$
Uniform		U	$y, 0 \leq y \leq 1$
Shifted exponential	A 5.5	SE	$1 - \exp(-y), 0 \leq y \leq \infty$
Shifted Rayleigh	A 5.13	SR	$1 - \exp\left(-\frac{1}{2}y^2\right), 0 \leq y \leq \infty$
Extreme value I largest (Gumbel)	A 5.11	G	$\exp[-\exp(-y)]$
Extreme value I smallest	A 5.11	EVIS	$1 - \exp[-\exp(y)]$
Type 2			
Lognormal	A 5.9	LN	
Gamma	A 5.6	GM	
Extreme value II largest	A 5.12	EVIIL	
Extreme value III smallest (Weibull)	A 5.13	W	

Table B.2 Coefficients in formula for $R = \rho'_{ij} / \rho_{ij}$ for selected distributions.

		Coefficients					
X_j	X_i	a	b	c	d	e	f
N	N	1					
	SE	1.107					
	SR	1.014					
	G	1.031					
	LN	*note 1*					
	GM	1.001	−0.007	0.118			
	W	1.031	−0.195	0.328			
SE	SE	1.229			−0.367	0.153	
	SR	1.123			−0.100	0.021	
	G	1.142			−0.154	0.031	
	LN	1.098	0.019	0.303	0.003	0.025	−0.437
	GM	1.104	−0.008	0.173	0.003	0.014	−0.296
	W	1.147	0.145	0.010	−0.271	0.459	−0.467
SR	SR	1.028			−0.029	0	
	G	1.046			−0.045	0.006	
	LN	1.011	0.014	0.231	0.001	0.004	−0.130
	GM	1.014	−0.007	0.126	0.001	0.002	−0.090
	W	1.047	−0.212	0.353	0.042	0	−0.136
G	G	1.064			−0.069	0.005	
	LN	1.029	0.014	0.233	0.001	0.004	−0.197
	GM	1.031	−0.007	0.131	0.001	0.003	−0.132
	W	1.064	−0.210	0.356	0.065	0.003	−0.211

note 1: $= V_j / [\ln(1 + V_j^2)]^{1/2}$

B.3 Orthogonal Transformation of Normal Random Variables

Let **X** be a correlated vector of basic variables, with mean

$$E(X) = [E(X_1), E(X_2), \ldots, E(X_n)] \tag{B.11}$$

and covariance matrix (*cf*. A.123)

$$\mathbf{C}_X = \mathrm{cov}(X_i, X_j)_{n\times m} = (\sigma_{ij})_{n\times m} \tag{B.12}$$

Table B.3 Coefficients on formula for $R = \rho'_{ij} / \rho_{ij}$ for selected Type 2 distributions.

X_j	X_i	Coefficients a	b	c	d	e	f	g	h	k	l	Maximum Error
LN	LN	*note 1*										
	GM	1.001	0.004	0.223	0.033	0.002	−0.104	−0.016	0.130	−0.119	0.029	4.0%
	W	1.031	0.052	0.220	0.052	0.002	0.005	−0.210	0.350	−0.174	0.009	2.4%
GM	GM	1.002	−0.012	0.125	0.022	0.001	−0.077	−0.012	0.125	−0.077	0.014	4.0%
	W	1.032	−0.007	0.121	0.034	0	−0.006	−0.202	0.339	−0.111	0.003	4.0%
W	W	1.063	−0.200	0.337	−0.004	−0.001	0.007	−0.200	0.337	0.007	−0.007	2.6%

note 1: $= \ln(1+\rho V_i V_j) / [\rho\sqrt{\ln(1+V_i^2)\cdot\ln(1+V_j^2)}]$

with $\mathrm{cov}(X_i, X_i) = \mathrm{var}(X_i)$. The covariance matrix will be strictly diagonal if the $\mathbf{X}$ are uncorrelated. Recall that 'correlation' is a measure of linear dependence (see Section A.7.3). This is a sufficient measure of dependence for Normal distributions.

An uncorrelated vector $\mathbf{U}$, and a linear transformation matrix $\mathbf{A}$, is now sought, such that

$$\mathbf{U} = \mathbf{AX} \quad \text{or} \quad \mathbf{A}^{-1}\mathbf{U} = \mathbf{X} \tag{B.13}$$

It is clearly desirable that the transformation (B.13) is also orthogonal, i.e. the vector represented by $\mathbf{X}$ is unchanged in length under the transformation $\mathbf{A}$. From well-known matrix theory this implies that $\mathbf{A}^T = \mathbf{A}^{-1}$. Further, the vector $\mathbf{U}$ will be uncorrelated if there exists a covariance matrix $\mathbf{C}_U$ which is strictly diagonal.

Under the linear transformation (B.13) the covariance matrix (B.12) is also transformed. It becomes the covariance matrix $\mathbf{C}_U$ for $\mathbf{U}$:

$$\begin{aligned}\mathbf{C}_U &= \mathrm{cov}(U_i, U_j) = \mathrm{cov}(\mathbf{U}, \mathbf{U}^T)\\ &= \mathrm{cov}(\mathbf{AX}, \mathbf{AX}) = \mathrm{cov}(\mathbf{AX}, \mathbf{X}^T\mathbf{A}^T)\\ &= \mathbf{A}\ \mathrm{cov}(\mathbf{X}, \mathbf{X}^T)\mathbf{A}^T\end{aligned} \tag{B.14}$$

or

$$\mathbf{C}_U = \mathbf{AC}_X\mathbf{A}^T \tag{B.15}$$

To obtain **U** as an uncorrelated vector, the matrix **A** which makes (B.15) diagonal is sought. Thus, it is required that the off-diagonal terms (i.e. $i \neq j$) in $\mathbf{C}_U$ be zero. This can be done by finding the characteristic values (eigen-values) of $\mathbf{C}_X$ using well-known methods. In particular, consider a matrix **D**, having only diagonal coefficients λ_{ii}, defined by

$$\mathbf{C}_X\mathbf{A} = \mathbf{A}\mathbf{D} \quad \text{or} \quad \mathbf{C}_X = \mathbf{A}\mathbf{D}\mathbf{A}^T \tag{B.16}$$

(provided that **A** is non-singular). Equation (B.16) may be written as a system of linear equations

$$\sum_i c_{ij}\ a_{jk} = a_{ik}\,\lambda_{ii}, \qquad i, k = 1, 2, \dots, n \tag{B.17}$$

Equation (B.17) represents a system of n linear equations with $j = 1, \dots, n$ terms of unknown coefficients a_{ij} on the left-hand side; typically

$$\begin{aligned} c_{11}\ a_{11} + c_{12}\ a_{21} + c_{13}\ a_{31} + \cdots &= a_{11}\ \lambda_{11} \\ c_{11}\ a_{12} + c_{12}\ a_{22} + c_{13}\ a_{32} + \cdots &= a_{12}\ \lambda_{22} \\ &\text{etc.} \end{aligned} \tag{B.18}$$

This system of equations is homogenous and may be written using the Kronecker delta $\delta_{ij}(= 1 \text{ if } i = j, \text{ otherwise} = 0)$:

$$\sum_i (c_{ij} - \lambda_{ii}\delta_{ij})a_{jk} = 0, \qquad i, k = 1, \dots, n \tag{B.19}$$

A system such as equations (B.19) has a non-trivial solution only if, for any value of k,

$$|c_{ij} - \lambda_{ii}\delta_{ij}| = 0 \tag{B.20}$$

or, more generally,

$$|\mathbf{C}_X - \lambda \mathrm{I}| = 0 \tag{B.21}$$

where **I** is the identity matrix. Equation (B.20) is the well-known 'characteristic' equation. Its solution, the 'characteristic values' $\lambda_{ii}\quad i = 1, \dots, n$, of matrix **D** are obtained by expanding (B.20) and solving the determinant

$$\left|\begin{matrix} c_{11} - \lambda_{11} & c_{12} & c_{13} \\ c_{21} & c_2 - \lambda_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} - \lambda_{33} \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \end{matrix}\right|_{n \times n} = 0 \tag{B.22}$$

There are standard procedures available to obtain λ.

For each λ_{ii}, Equations (B.18) have a solution $(a_{i1}, a_{i2}, \dots, a_{ik}, \dots, a_{in})$. This is termed the 'characteristic vector' and yields the ith row of matrix **A**. It represents the components of one of the uncorrelated vectors U_i in terms of the correlated vectors X_i, (and vice versa since **A** is symmetric). Thus solution for all characteristic values λ_{ii} and hence the characteristic vectors produces the complete matrix **A**.

With **A** known, the uncorrelated vector **U** is then defined by

$$\mathbf{U} = \mathbf{AX} \tag{B.13}$$

with

$$E(\mathbf{U}) = \mathbf{A}E(\mathbf{X}) \tag{B.23}$$

and using (B.15) there is obtained

$$\mathbf{C}_U^{1/2} = (\mathbf{AC}_X\mathbf{A}^T)^{1/2} \tag{B.24}$$

where $C_U^{1/2}$ is the matrix of standard deviations of **U**. This comes as about as follows. Since **U** in uncorrelated, $\mathbf{C}_U$ is strictly diagonal. For a diagonal matrix, such as **D**, say, it is well known that $\{D_{\text{ij}}\}^2 = \{D_{\text{ij}}{}^2\}$. Hence $C_U^{1/2}$ consists only of terms $C_U^{1/2} = \sigma_{u_i} = \lambda_{ii}^{1/2}$ along the leading diagonal, where the λ_{ii} are the characteristic values.

A special case arises if the vector **U** consists of only one term, **Z**, say. Then (B14) is given by

$$\mathbf{Z} = \mathbf{AX} \tag{B.25}$$

where the matrix **A** has become just one row of coefficients (see Sections 4.2 and A.11.1). The covariance matrix for **Z** is then one term (the variance) given by

$$\mathbf{C}_Z = \text{var}(\mathbf{Z}) = \mathbf{AC}_X\mathbf{A}^T \tag{B.26}$$

Mathematical aside. The standard deviations $\lambda_{ii}^{1/2}$ of $\mathbf{C}_U$ must all be positive (or zero), since they have no physical meaning otherwise. This means that the transformation (B.15) must have special properties; in particular it is said that $\mathbf{C}_X$ must be a positive (semi-) definite matrix. In matrix terminology, this means that the determinant of any minor,

$$\det M_i = \begin{vmatrix} c_{11} & \cdot & \cdot & c_{i1} \\ \vdots & & & \vdots \\ c_{1i} & \cdot & \cdot & c_{ii} \end{vmatrix} \tag{B.27}$$

is defined as positive definite if $\det M_i > 0$ or positive semi-definite if $\det M_i \geq 0$. A matrix is non-negative definite if it is either of these.

B.4 Generation of Dependent Random Vectors

In some situations it may be necessary to generate dependent variables. As will be seen, with the exception of Normal distributed variables, this may not be easy in practice.

If the random variables in vector **X** are independent of each other, the joint probability density function can be decomposed as (*cf*. A.117)

$$f_{\mathbf{X}}(x) = \prod_{i=1}^{n} f_{X_i}(x_i) \tag{B.28}$$

where $f_{X_i}(x_i)$ is the marginal probability density function of the random variable X_i. The generation of the vector **X** follows directly from the inverse transform method (see Section **3.4.3**) for each random variable X_i separately.

If there is dependence between the random variables constituting **X**, (B.28) is replaced by (cf. (A.116) and A.3)

$$f_{\mathbf{X}}(\mathbf{x}) = f_{X_1}(x_1) f_{X_2|X_1}(x_2|x_1) \ldots\ f_{X_n|X_1,\ldots,X_{n-1}}(x_n|x_1,\ldots,x_{n-1}) \tag{B.29}$$

where $f_{X_k|X_i,\ldots,X_n}$ is the conditional probability density function of X_k given that $X_1 = x_1, \ldots$, and where $f_{X_1}(x_1)$ is the marginal probability density function of X_1.

Provided that the joint probability density function $f_{\mathbf{X}}(\mathbf{x})$ is known in terms of the conditional probability density functions as in (B.29), the inverse transform method of Section 3.3.3 can be extended to generate a sample vector $\hat{\mathbf{x}}$, such that $\hat{\mathbf{x}}$ is drawn from a distribution with mean vector $\mu_{\mathbf{X}}$ and covariance matrix $\mathbf{C}_X$ as defined by the joint probability density function $f_{\mathbf{x}}(\,)$ for the multivariate normal distribution (see also Appendix C):

$$f_{\mathbf{X}}(\mathbf{x}, \mathbf{C}_X) = \frac{1}{(2\pi)^{n/2}|\mathbf{C}_X|^{1/2}} \exp\left[-\frac{1}{2}(\mathbf{x} - \mu_{\mathbf{X}})^T \mathbf{C}_X^{-1}(\mathbf{x} - \mu_{\mathbf{X}})\right] \tag{B.30}$$

The vector **X** of correlated Normal (Gaussian) distributed random variables with given covariance matrix $\mathbf{C}_X$ can be generated from a vector of independent standardised Normal random variables such as $\mathbf{Y} = \{Y_i\}$ using either the orthogonal transformation of Section B.3 or the Rosenblatt transformation of Section B.1.

To be specific, let the n correlated random variables be **X**, with known mean vector $\mu_{\mathbf{x}}$, and known covariance matrix $\mathbf{C}_X$. Further, let **U** be the vector of independent random variables with strictly diagonal covariance matrix $\mathbf{C}_U$. Then the sample (i.e. generated) dependent values of **X** can be obtained from the sampled independent **U** by the orthogonal transformation (*cf*. B.14):

$$\mathbf{X} = \mathbf{A}^T \mathbf{U} \tag{B.31}$$

where, since $\mathbf{C}_X$ is positive definite and symmetric, the orthogonal transformation matrix $\mathbf{A}^T = \mathbf{A}^{-1}$ can be defined through the transformation of the covariance matrix (*cf*. B.15):

$$\mathbf{C}_U = \mathbf{A}\mathbf{C}_X\mathbf{A}^T \quad \text{or} \quad \mathbf{C}_X = \mathbf{A}^T\mathbf{C}_U\mathbf{A} \tag{B.32}$$

If **B** is substituted for $\mathbf{A}^T = \mathbf{A}^{-1}$ then

$$\mathbf{C}_X = \mathbf{B}\mathbf{C}_U\mathbf{B}^T \tag{B.33}$$

The elements b_{ij} of the square matrix **B** are given by

$$c_{ij} = \sum_{k=1}^{n} b_{ik}\, u_{kk}\, b_{jk} \tag{B.34}$$

where c_{ij} is known from $\mathbf{C}_X$. Also, $\mathbf{U} = \mathbf{Y}$ is the vector of independent standardized Normal variables, $u_{kk} = 1$ for all k, $u_{jk} = 0$, $j \neq k$.

The transformation (B.31) then becomes, after allowing for the normalization of the mean of **X**,

$$\mathbf{X} - \mu_{\mathbf{X}} = \mathbf{BY} \quad \text{or} \quad \mathbf{X} = \mathbf{BY} + \mu_{\mathbf{X}} \tag{B.35}$$

from which the correlated variables **X** can be generated for samples of **Y**. Further (B.32) reduces to

$$\mathbf{C}_X = \mathbf{B}\mathbf{B}^T = \mathbf{A}^T\mathbf{A} \tag{B.36}$$

where the matrix **B** is square lower triangular and obtained directly from **A** as indicated for (B.33).

Alternatively, a recursive formula may be derived to obtain the elements b_{ij} of **B** (*cf.* Rosenblatt, 1952). Noting that **B** is lower triangular, consider for instance the following example of (B.36):

$$\begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix} = \begin{bmatrix} b_{11} & 0 & 0 \\ b_{21} & b_{22} & 0 \\ b_{31} & b_{32} & b_{33} \end{bmatrix} \begin{bmatrix} b_{11} & b_{21} & b_{31} \\ 0 & b_{22} & b_{32} \\ 0 & 0 & b_{33} \end{bmatrix} \tag{B.37}$$

from which

$$\sigma_{11} = b_{11}^2 \tag{B.38a}$$
$$\sigma_{22} = b_{21}^2 + b_{22}^2 \tag{B.38b}$$
$$\sigma_{33} = b_{31}^2 + b_{32}^2 + b_{33}^2 \tag{B.38c}$$

or generalizing

$$\sigma_{ii} = \sum_{k=1}^{n} b_{ik}^2 \quad k \le i$$

and

$$\sigma_{21} = \sigma_{12} = b_{21}b_{11} \tag{B.38d}$$
$$\sigma_{31} = \sigma_{13} = b_{31}b_{11} \tag{B.38e}$$
$$\sigma_{32} = \sigma_{23} = b_{31}b_{21} + b_{32}b_{22} \tag{B.38f}$$

or generalizing

$$\sigma_{ij} = \sum_{k=1}^{n} b_{ik}b_{jk} \quad k \le j < i$$

The recursive result now follows directly from (B.38a):

$$b_{11} = \sigma_{11}^{1/2}$$

from (B.38d)

$$b_{21} = \frac{\sigma_{21}}{b_{11}} = \frac{\sigma_{21}}{\sigma_{11}^{1/2}}$$

from (B.38b)

$$b_{22}^2 = \sigma_{22} - b_{21}^2$$

i.e.

$$b_{22} = \left(\sigma_{22} - \frac{\sigma_{21}^2}{\sigma_{11}}\right)^{1/2}$$

from (B.38e)

$$b_{31} = \frac{\sigma_{31}}{b_{11}} = \frac{\sigma_{31}}{\sigma_{11}^{1/2}}$$

from (B.38f)

$$b_{32} = \frac{\sigma_{32} - b_{31}b_{21}}{b_{22}}$$

etc. From thus it can be readily verified that the recursive formula

$$b_{ij} = \frac{\sigma_{ij} - \sum_{k=1}^{j-1} b_{ik}b_{jk}}{\left(\sigma_{jj} - \sum_{k=1}^{j-1} b_{jk}^2\right)^{1/2}} \tag{B.39}$$

is valid provided that $\sum_{k=1}^{0}$ is interpreted as zero [Rubinstein, 1981].

Example B.1 It is desired to generate correlated Normal variates X_1 and X_2 having means 10 and 12, respectively, and a non-negative definite covariance matrix

$$\mathbf{C}_X = \begin{bmatrix} 2 & 2\sqrt{2} \\ 2\sqrt{2} & 7 \end{bmatrix}$$

From equations (B.38)

$$b_{11} = \sigma_{11}^{1/2} = \sqrt{2}$$

and

$$b_{21} = \frac{\sigma_{21}}{b_{11}} = \frac{2\sqrt{2}}{\sqrt{2}} = 2$$

and

$$b_{22} = \left(\sigma_{22} - \frac{\sigma_{21}^2}{\sigma_{11}}\right)^{1/2} = \left(7 - \frac{8}{2}\right)^{1/2} = \sqrt{3}$$

Hence

$$\mathbf{B} = \begin{bmatrix} \sqrt{2} & 0 \\ 0 & \sqrt{3} \end{bmatrix}$$

and, from (B.35),

$$X_1 = \sqrt{2}\, Y_1 + 10$$
$$X_2 = 2Y_1 + \sqrt{3}\, Y_2 + 12$$

where Y_1 and Y_2 are independent standardized normal variables (generated through a random number generator).